

Physics layer Φ (frozen)

self-supervised dynamics · behavior proposals

trained once, reused across taxonomies

embeddings
+ proposals



Semantic layer Ω (revisable)

anchor-guided pseudo-labeling · self-training

taxonomy evolves $\mathcal{Y} \rightarrow \mathcal{Y}'$

only Ω retrains

budget retention · own full-budget top-1 kept at 10% labels (canine: 9.8 vs 11.1% chance at 13% budget)

NTU60

90.7%

NTU120

88.9%

canine

28.9%

100%